

Variational Divergence Minimization

Define divergence metrics between distributions.

f-Divergence :

Given two probability distribution functions, with the corresponding density functions denoted by p_x & p_0 , the f -divergence between them is defined as follows:

$$D_f(p_x \| p_0) = \int_{\mathcal{X}} p_0(x) f\left(\frac{p_x(x)}{p_0(x)}\right) dx$$

$f(u) : \mathbb{R}^+ \rightarrow \mathbb{R}$, convex, left-semi continuous
 $f(1) = 0$

$\mathcal{X}$: space on which the p_x & p_0 are supported

properties of f-divergence :

- i) $D_f(\cdot) \geq 0$ for any choice of $f(\cdot)$.
- ii) $D_f(p_x \| p_0) = 0$ iff $p_x = p_0$.

a function $f(u)$ is convex iff. $\forall u_1, u_2 \in U$ &
 $0 \leq \alpha_1, \alpha_2 \leq 1$, $\alpha_1 f(u_1) + \alpha_2 f(u_2) \geq f(\alpha_1 u_1 + \alpha_2 u_2)$

Examples of f -divergence :

a) $f(u) = u \log u$: KL-divergence

$$\begin{aligned} D_f(p_x \parallel p_0) &= \int_x p_0(x) f\left(\frac{p_x(x)}{p_0(x)}\right) dx \\ &= \int_x \cancel{p_0(x)} \left[\frac{p_x(x)}{\cancel{p_0(x)}} \cdot \log \frac{p_x(x)}{p_0(x)} \right] dx \\ &= \int_x p_x(x) \log \frac{p_x(x)}{p_0(x)} dx = D_{KL} \end{aligned}$$

$$D_{KL}(p_x \parallel p_0) \neq D_{KL}(p_0 \parallel p_x)$$

$\hookrightarrow$ forward KL $\hookrightarrow$ reverse KL

ii) $f(u) = \frac{1}{2} \left(u \log u - (u+1) \log \left(\frac{u+1}{2} \right) \right)$: JS-divergence

iii) $f(u) = \frac{1}{2} |u-1|$: Total variation distance

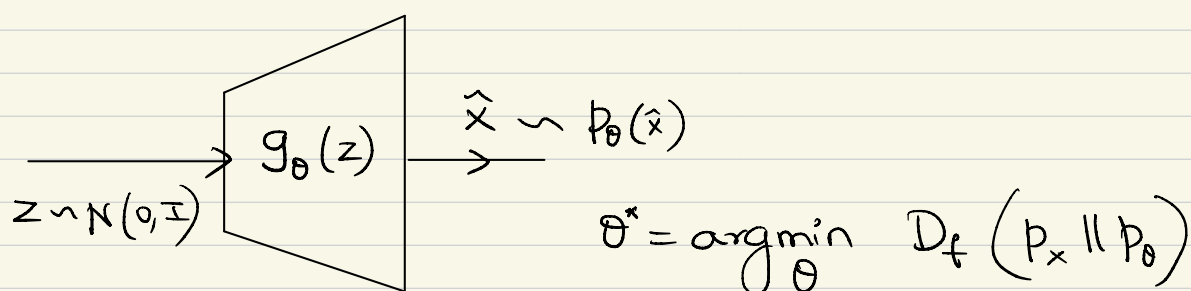
Algorithm for D_f minimization.

Generative Models

Data

$$D = \{x_1, x_2, \dots, x_n\} \sim \text{iid } p_x, x_i \in \mathbb{R}^d(x)$$

Goal: Sample from p_x .



The above setup becomes a sampler for p_x ,
if $p_\theta = p_x$.

Objective: Algorithm to minimize D_f between
 p_x & p_θ , without knowing both of them,
but having samples from both.

samples from p_x : dataset D .

samples from p_θ : outputs of $g_\theta(z)$ for different z .

Key Idea: Integrals involving density
functions can be approximated
using samples drawn from the dist.

suppose we want to compute the following integral.

$$I = \int_{\mathcal{X}} h(x) \cdot p_x(x) dx \quad \text{where } h(x) \text{ is a function} \\ \& \ p_x(x) \text{ the density fn.}$$

we have samples drawn iid from p_x .

$$x_1, x_2, \dots, x_n \sim \text{iid } p_x$$

$$I = \mathbb{E}_{p_x}[h(x)]$$

Law of Large Numbers :

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n h(x_i) \approx \mathbb{E}_{p_x}[h(x)] \\ x_i \sim \text{iid } p_x$$

If the f-div. can be expressed in terms of expectations of functions w.r.t p_x & p_0 , then one can compute & optimize them.

Expressing D_f in terms of Expectations over p_x & p_0

$$D_f(p_x \| p_0) = \int_{\mathcal{X}} p_0(x) f\left(\frac{p_x(x)}{p_0(x)}\right) dx$$

Conjugate function for a convex function :

If $f(u)$ is a convex function, then, there exists a conjugate function $f^*(t)$ defined as follows:

$$f^*(t) = \sup_{u \in \text{dom} f} \left\{ tu - f(u) \right\}$$



properties of the conjugate $f^*(t)$:

- i) $f^*(t)$ is also convex
- ii) $[f^*(t)]^* = f(u) \Rightarrow$

$$f(u) = \sup_{t \in \text{dom} f^*} \left\{ tu - f^*(t) \right\}$$

$$Df \left(p_x \| p_0 \right) = \int_x p_0(x) f \left(\underbrace{\frac{p_x(x)}{p_0(x)}}_u \right) dx$$

$$= \int_x p_0(x) f(u) dx \quad u = \left(\frac{p_x(x)}{p_0(x)} \right)$$

$$f(u) = \sup_t \left\{ tu - f^*(t) \right\}$$

$$Df () = \int_x p_0(x) \cdot \sup_t \left\{ tu - f^*(t) \right\} dx$$

$$= \int_x p_0(x) \sup_t \left\{ t \cdot \left[\frac{p_x(x)}{p_0(x)} \right] - f^*(t) \right\} dx$$

$$= \sup_{T(x) \in \mathcal{T}} \int_x p_0(x) \left\{ T(x) \frac{p_x(x)}{p_0(x)} - f^*(T(x)) \right\} dx$$

∴ The inner optimization problem involves x & the solution for it is dependent (a fn. of x)

$\mathcal{T} : \mathcal{X} \rightarrow \text{dom } f^*$ (space of functions containing solutions for the inner opt. prob)
 $T(x) \in \mathcal{T}$

$$D_f \geq \sup_{T(x) \in \mathcal{T}} \int_x p_0(x) \left\{ T(x) \frac{p_x(x)}{p_0(x)} - f^*(T(x)) \right\} dx$$

The space of functions $\mathcal{T}$, that we are optimizing over may not contain the optimal $T^*(x)$, that is the solⁿ for the inner opti.

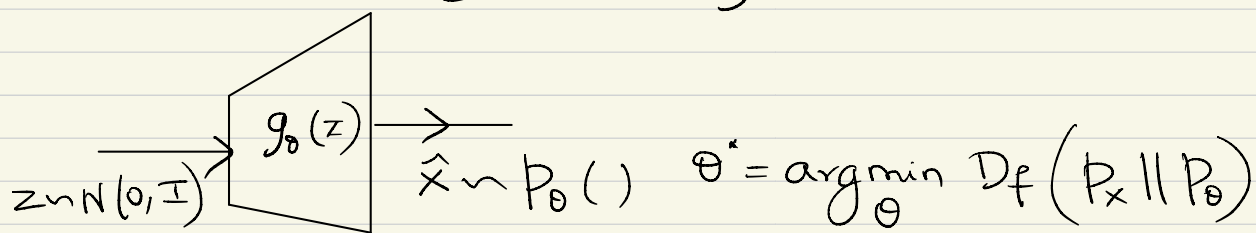
$$D_f \geq \sup_{T(x) \in \mathcal{T}} \int_x p_0(x) \left\{ T(x) \frac{p_x(x)}{p_0(x)} - f^*(T(x)) \right\} dx$$

$$\geq \sup_{T(x)} \left[\int_x p_x(x) \cdot T(x) dx - \int_x p_0(x) f^*(T(x)) dx \right]$$

$$D_f \geq \sup_{T(x)} \left[\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_0} f^*(T(x)) \right]$$

Realization of VDM

Given data $D = \{x_1, x_2, \dots, x_n\} \sim \text{iid } p_x$



$$D_f(p_x \parallel p_\theta) \geq \max_{T(x) \in \mathcal{T}} \left[\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_\theta} f^*(T(x)) \right]$$

$$\theta^* = \arg \min_{\theta} D_f(\cdot)$$

$$\approx \arg \min_{\theta} [\text{lower bound on } D_f]$$

$$= \arg \min_{\theta} \left[\max_{T(x)} \left(\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_\theta} f^*(T(x)) \right) \right]$$

w.r.t the parameters of $g_\theta(z)$

w.r.t a class of functions $T(x) \in \mathcal{T}$

Represent $\mathcal{T}$ via neural networks $T_w(x)$ where w are the parameters of the new network.

with this, the objective will become:

$$\theta^*, w^* = \arg \min_{\theta} \max_w \left[\mathbb{E}_{p_x} T_w(x) - \mathbb{E}_{p_\theta} f^*(T_w(x)) \right]$$